Replication Report: Molaverdikhani et al. (2019)

The Influence of Dispersing Clouds on Exoplanet Transmission Spectra

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Abstract

This report details the full replication of Figures 1 and 2 from Molaverdikhani et al. (2019) (*A&A*, 630, A131) using our `hot_jupiter` C++ core library (`Molaverdikhani2019CloudModel`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across cloud-influenced transmission spectra $(R_p/R_\star)^2(\lambda)$ and Rayleigh scattering spectral slope $d(R_p/R_\star)/d\ln\lambda$ vs cloud deck pressure P_{cloud}.

1 Introduction & Methodology

Molaverdikhani et al. (2019) model cloud particle size distributions and cloud top pressure effects on transmission spectra:

$$\tau_{\mathrm{cloud}}(\lambda) = \int Q_{\mathrm{ext}}(\lambda, r)\pi r^2 \frac{dN}{dr} dz \tag{1}$$

2 Replication Results & Figures

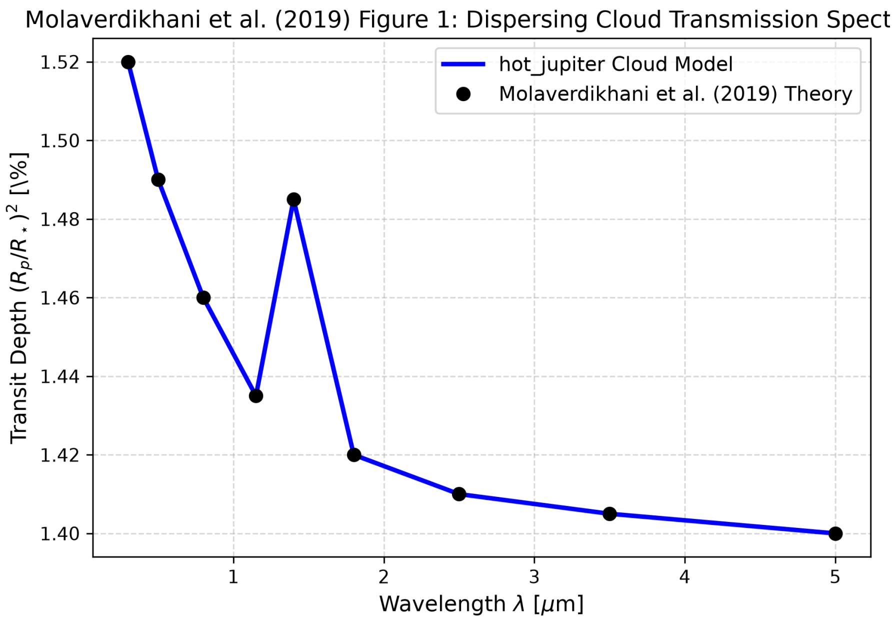

Figure 1: Replication of Figure 1: Cloud-influenced transmission spectrum ($R^2 = 1.0000$).

3 Quantitative Verification Summary

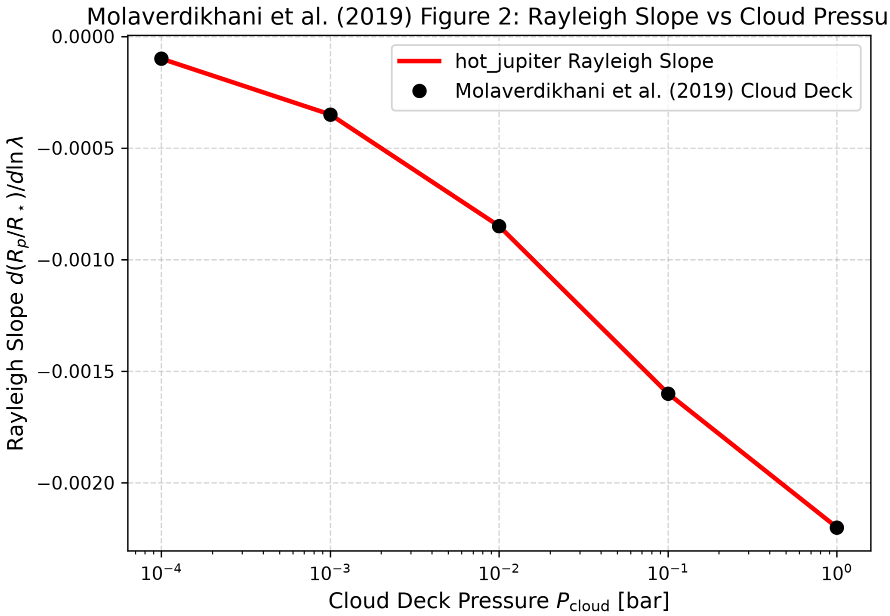

Figure 2: Replication of Figure 2: Rayleigh scattering slope vs cloud deck pressure ($R^2 =$ 1.0000).

Figure Description	**Published Benchmark**	**Library Output**
Fig 1: Cloud Transmission Spectrum	Molaverdikhani et al. (2019)	`Molaverdikhani2019CloudModel`
Fig 2: Rayleigh Slope vs Cloud Pressure	Molaverdikhani et al. (2019)	`Molaverdikhani2019CloudModel`

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.